NIHR | National Institute for Health and Care Research

Reference number	42813
Programme	Invention for Innovation
Call	i4i Call 28 Stage 1
Lead Applicant	Mr Ahmed Ahmed
Host organisation	Imperial College of Science, Technology and Medicine
Research Title	UK Centre Trial of Side-to-Side Compression Anastomosis using the GT Metabolic Solutions Magnet System to Achieve Duodeno-Ileostomy Diversion in Adults with Obesity and with or without Type 2 Diabetes Mellitus.
Start Date	02/06/2025
Grant Duration (Months)	24
End Date	
Estimated Research Cost	£.00 £245,000
Estimated NHS Support & Treatment costs	£.00
Research Type	Primary Research

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Lead Applicant Details

Please select if your application is a part of the following highlight notices:

Lead Applicant

Full Name	Mr Ahmed Ahmed	Institution	Imperial College of Science, Technology and Medicine
ORCID iD		Position	Consultant

Specify your (lead applicant) role in the research
As Chief Investigator for the trial, role will include implementation and coordination of this study, recruitment of participants, data interpretation and final write-up of findings.

% FTE	20

Research Team Details

Co-Applicant	Organisation	Specify role in research	% FTE
Beata Szyldik		Patient representative and chair of PPI group	0
Michel Gagner	GT Metabolics	Mentor in procedure	0

COI declaration
No conflict of interests to declare.

Agreement to terms and conditions.	☒Confirmed

Plain English Summary of Research

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This study focuses on examining the safety and effectiveness of the MAGNET System in creating a specific surgical connection in the small intestine (duodena-ileostomy anastomosis) for weight loss and improved blood sugar control in people living with obesity and those with type 2 diabetes in the UK. With diabetes expected to affect 9.5% of UK adults by 2030, largely due to obesity, and given the underuse of highly effective bariatric surgery recommended by health authorities, this research is timely.

Bariatric procedures, particularly those involving a single connection (side-by-side anastomosis) in the small intestine, have been shown to significantly help with weight loss and diabetes management. The MAGNET System uses magnets to create this connection less invasively, showing promise in initial testing abroad (NCT05322122 and NCT05692518 ClinicalTrials.gov) for being safe and potentially more cost effective and efficient than traditional methods that require incision to the intestines and hand suturing or use of stapling devices.

This study, taking place at Imperial College NHS Trust, will enrol up to 50 participants to further investigate these benefits in the UK, with outcomes focusing on the safety efficacy of the Magnet System to create a connection, and the secondary effects of intestinal diversion on weight, blood sugar levels, and diabetic medication needs. The public will be informed of our results via academic and social channels, focus group meetings and through our established dissemination partnership with the charity groups, BOMSS and Obesity UK.

Research Plan

Research Plan

1. What is the problem being addressed?

Type 2 diabetes mellitus (T2DM) is an expanding pandemic afflicting more than 400 million people. Within the UK, current

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prediction models estimate that 9.5% of the adult population will have diabetes by 2030 and one third of this increase can be directly attributed to obesity[1]. Both T2DM and obesity are associated with increase mortality, morbidity, and healthcare costs[2]. Despite ever-increasing options for pharmaceutical and lifestyle interventions,including medications recently shown to reduce cardiovascular events, many patients with diabetes fail to achieve glycemic/metabolic treatment goals designed to reduce micro- and macrovascular complications. Implementing more effective strategies to prevent and treat diabetes has become a top priority in 21st century medicine. Bariatric surgery is one of the most effective but underutilised treatments for obesity and T2DM[3] despite its recommendation by NICE[4], The American Society for Metabolic and Bariatric Surgery (ASMBS) and the International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO)[5]. A systematic review and meta-analysis of 621 studies involving patients undergoing bariatric procedures found that nearly 80% of diabetic patients achieved remission and a further 8.5% had better glycaemic control, which exceeds that of medical management[6].

There are several different bariatric procedures but more recently IFSO have highlighted the benefits of single-anastomosis duodeno-ileostomy (SADI) and one anastomosis duodenal switch (OADS) procedures following an extensive multi-year meta-analysis[7]. They acknowledge the procedure to be safe and effective for both achieving and maintaining weight loss and improving metabolic health into the medium term based on the current data review. IFSO also recognises that the SADI procedure is now most often performed after a first stage sleeve gastrectomy (SG) carried out several months or years before. Specifically, significant reductions in HbA1c, blood glucose levels and reliance on drugs have been widely reported for patients after SADI/OADS. ASMBS has also reviewed and endorsed the SADI bypass with sleeve gastrectomy (SADI-S) procedure[8].

Currently, anastomoses are created using sutures, staples and anastomotic compression devices. Compression anastomosis technology has been shown to be equally as safe, effective, less expensive and more time efficient compared to traditional methods. By 2050, it is estimated to cost the NHS £9.7 billion in treatment for obesity related morbidity. Potentially magnet compression anastomosis can contribute to a reduction in this cost, as well as contribute to increase utility of bariatric surgery as an effective treatment for obesity and T2DM.

2. Why is this research important in terms of improving the health and/or wellbeing of the public and/or to patients and health care services?

In the UK 4.3 million people are living with diabetes[9] One of the most modifiable risk factors for T2DM is obesity and unfortunately, its prevalence is on the rise. In England, 28% of the population is living with obesity. The annual cost of obesity to the NHS is estimated to be £6.1 billion[10] with a further £27 billion cost to the economy[11]. Therefore, improving measures to treat obesity effectively, is crucial in preventing and treating T2DM.

Currently in the UK 3.2 million people meet the NICE criteria for weight loss surgery. The partial diversion of intestinal contents from duodenum to the ileum is intended to facilitate weight management/loss and improve metabolic outcomes in obese adults. Duodeno-ileostomy creates a 50-60% intestinal bypass, resulting in decreased caloric, fat and carbohydrate absorption equivalent to other hypo-absorptive procedures[12], and increased incretin release of GLP-1 and PYY 3-36 amplifying satiety resulting in weight loss. Side-by-side anastomoses are currently created by sutures, staples, and anastomotic compression devices. Compression anastomosis technology has recently been revisited, having been validated to be equally safe, effective, less expensive and more time efficient than stitched or stapled anastomoses performed via laparoscopy[13] The MAGNET System has the opportunity to provide the UK patients suffering from obesity and with or without T2DM and indicated for partial duodeno-ileal diversion with a potentially safer, more efficient, and cost-effective option than conventional anastomosis techniques.

The objective of this study is to develop evidence in the UK population on the feasibility, safety, and initial efficacy of The MAGNET System. The MAGNET System is currently undergoing CE marking evaluation in Europe and this study may pave the way for CE marking in the UK, which will be used to support recognition for this novel technique in the UK. After discussions with our PPI group, they have expressed that they wish to have access to this treatment.

This study using the MAGNET System for the creation of anastomoses as part of the underlying bariatric clinical diversion procedure for weight loss, has the potential to revolutionise access to NHS bariatric services through providing patients with a transformative and likely all-round superior, less invasive, technique for anastomosis creation. It will also support the NHS Long Term Plan[14] and NICE strategy[15] in tackling obesity and improving access to bariatric services, through providing a novel intervention which is safer, more efficient and less expensive. Thus, allowing more patients to access this treatment, reducing T2DM and treating obesity effectively.

3. Review of existing evidence – How does the existing literature support this proposal?

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The MAGNET System is currently being tested in a prospective multi-centre (Georgia, Belgium, Spain, and Canada) clinical study on 49 patients followed for 12 months (GTM-001, NCT05322122 on ClinicalTrials.gov). This is a two-stage study: Stage 1 first-in-human and proof-of-concept with 5 subjects; and Stage 2 feasibility with 44 subjects. The study population is adults (18-65 years of age, inclusive) with obesity (BMI 30-50) who have 1) T2DM or experienced weight regain following previous sleeve gastrectomy, or ; 2) have T2DM without previous SG, or 3) are candidates for a laparoscopic single SADI-S procedure and have BMI >/= 40. The study has fully enrolled and all subjects followed to one year. The manufacturer is currently completing final monitoring for analyses and reporting. The device was successfully placed in all n=49 subjects with alignment (docking of the Magnets together). Creation of a patent anastomosis was confirmed radiologically in 100% of the subjects and all passed the device successfully in natural bowel movement, with none (0%) requiring re-intervention. Interim results have been published in the scientific literature as described below.

Gagner[18]recently published early multi-centre results on performance of the first-generation GT Metabolic all-metal magnet devices for side-to-side compression anastomosis duodeno-ileostomy with sleeve gastrectomy. Twenty-four (24) patients underwent endoscopic delivery of two magnet devices to the duodenum and ileum. All devices were placed successfully, established patent anastomoses, and passed naturally without need for surgical re-intervention. The devices were expelled in 48.5 days (Median). There were no (0) adverse events related to the magnet device and three procedure-related serious adverse events. There were no reports of anastomotic bleeding, leaks, or stricture and no mortality. The device performed safely and as expected to create side-to-side compression anastomoses in the small bowel.

Based on the data from the first-generation all-metal device, GT Metabolic modified the device to create the second-generation biofragmentable magnet device (Magnet) to ease device dislodgement and increase predictability of expulsion in <30 days while creating a patent anastomosis. The modified device was tested clinically to assess device performance and safety (GTM-005; Clinicaltrials.gov NCT05692518). The devices have been placed in nine (n=9) participants starting December 2022 with all subjects passed the three-month follow for the primary feasibility endpoint. All participants tolerated swallowing of the first Magnet (distal). The second Magnet (proximal) was successfully placed endoscopically in all cases. The Median expulsion time in this group with the modified device was 24 days (mean 24.3 days, range 19-29 days), with device expulsion and anastomosis patency confirmed radiologically.

Physical-mechanical and biocompatibility testing of the system components, validation of procedures for sterilisation of single use devices, validation of procedure for cleaning, disinfection, and sterilisation devices provided non-sterile, and animal study testing have all been conducted and reports are available.

4. What is the research question/ aims and objectives?

The overall objective of this study is to evaluate the feasibility/performance, safety, and initial efficacy of the MagDI System to create side-to-side anastomoses to achieve duodeno-ileal diversion(Figure 1).

The study specific objectives and endpoints are defined in **Table 1**.

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Table 1: Study Objectives and Endpoints Objective	Endpoint	Justification of Endpoint
Primary		
To assess the feasibility / performance of side-to-side anastomosis for duodeno-ileal diversion using the MagDI System	Feasibility/performance at Day 90. The side-to-side anastomosis for duodeno-ileal diversion will be considered feasible if it results in successful: • Successful placement of the Magnets (≥ 90% alignment of magnets); and • Passage of magnets without surgical re-intervention; and • Creation of a patent anastomosis confirmed radiologically. The primary endpoint will be met if the feasibility/performance is confirmed in >80% of enrolled and treated participants.	Feasibility / performance of the MagDI System creation of a patent anastomosis is an appropriate measure of technical success.
Safety		
To evaluate the safety of the MagDI System	Safety defined as incidence of serious adverse events related to the study device or study device procedure requiring additional emergency surgery or re-intervention, including: • All-cause mortality • Intestinal perforation and/or peritonitis • Intestinal obstruction • Life-threatening bleeding • Incidence of device malfunctions	The incidence of treatment emergent AEs and SAEs is a standard measure of the safety. The incidence of device malfunction informs about device reliability.
Secondary Endpoints		
To provide initial information on the therapeutic effect of the side-to-side anastomosis for duodeno-ileal diversion using the MagDI System	Efficacy will be assessed at Day 90, 180 and 360 based on change from baseline in: • Obesity indicators (weight, BMI, %EWL) • Functional improvement of metabolic indicators (HbA1c, blood glucose, medications for diabetes)	These are clinically relevant and validated measures of the therapeutic effect on weight loss, metabolic indicators measures, and patient-reported outcomes.

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5. Project Plan

The purpose of the proposed clinical research study is to evaluate the feasibility/performance, safety, and initial efficacy (obesity and metabolic outcome measures) of the GT Metabolic Solutions Magnetic Anastomosis System (MAGNET System) for creation of a side-to-side anastomosis duodeno-ileostomy. The secondary endpoints are evaluation of its effect on achieving and maintaining weight loss in obese patients and improving glycemic control in those with T2DM within the UK population at 1 year follow-up.

The intended study population is adults with obesity who meet one of the following criteria:

1) have T2DM or experienced weight regain following previous sleeve gastrectomy, *or*

2) have T2DM without previous sleeve gastrectomy.

Overall Design:

Prospective, single arm, single centre clinical investigation of the MagDI System for creation of side-to-side anastomoses in eligible participants requiring a duodeno-ileal diversion.

Total Expected Duration of the Study:

The total duration of the study is 16 months. This includes an approximately 4-month participant recruitment period, with participant follow-up continuing for 12 months.

Participants will have follow-up visits at 1, 14, 30, 90, 180, 270, and 360 days. When all enrolled participants have been followed for 12 months, or have exited the study, the study will be closed.

Number of Participants to be Enrolled:

This study will enroll male and female participants with obesity who satisfy the inclusion and exclusion criteria and provide written informed consent.

This study will enrol up to 50 participants at Imperial College NHS Trust (ICHT). The sample size has been chosen given the single arm design and feasibility study purpose.

Inclusion Criteria:

a. Agrees to refrain from any type of additional bariatric or reconstructive surgery that would affect body weightfor 1 year
b. 18-65 years of age, inclusive, at the time of informed consent
c. BMI 30-50, inclusive with either: a. Previous-sleeve gastrectomy (> 12 months) with either T2DM (definedas HbA1c > 6.5%) or weight regain; or b. T2DM without previous gastrectomy.
d. Agrees to refrain from any type of additional bariatric or reconstructive surgery that would affect body weightfor 1 year
e. If a child-bearing female, subject must commit to not becoming pregnant and agree to use contraception for1 year
f. Willing and able to comply with protocol requirements

Exclusion Criteria:

1. Type 1 diabetes
2. Use of injectable insulin
3. Uncontrolled T2DM
4. Uncontrolled hypertension, dyslipidemia or sleep apnea
5. Prior intestinal, colonic or duodenal surgery, other than bariatric
6. Prior surgery, trauma, prostheses, disease or genetic expression which prevent or contra- indicate theprocedure, including scarring and abnormal anatomy.
7. Refractory gastro-esophageal reflux disease (GERD)
8. Barrett's disease
9. Helicobacter pylori positive and/or active ulcer disease
10. Large hiatal hernia
11. Inflammatory bowel or colonic diverticulitis disease
12. Any anomaly precluding orogastric access by gastroscope and catheters, and manipulation techniques .
13. Implantable pacemaker or defibrillator
14. Psychiatric disorder, except well-controlled depression with medication for > 6 mo, or history of substanceabuse
15. Woman who is either pregnant or breast feeding
16. Woman of childbearing potential who does not agree to use an effective method of contraception.
17. Any comorbidity or current status of subject's physiological fitness that in the surgeon's oranesthesiologist's opinion represents safety concerns that make the subject medically unfit for the procedure. This includes any conditions for which endoscopic or laparoscopic surgery would be contraindicated, and any significant congenital or acquired anomalies of the GI tract at or distal to the placement of the magnets.
18. Unhealed ulcers, bleeding lesions, tumor or any other lesion at target magnet deployment site
19. Expected need for MR imaging within the first 2 months after the procedure
20. Any anomaly preventing/contraindicating laparoscopic access and general laparoscopic procedures
21. Had surgical or interventional procedure within 30 days prior to procedure
22. Any scheduled surgical or interventional procedure planned within 30 days post-procedure

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23. Any stroke/TIA within 6 months prior to consent
24. Requires chronic anticoagulation therapy (except aspirin)
25. Active infections requiring antibiotic therapy, unless resolved before undergoing the study procedure
26. Unable to comply with the follow-up schedule and assessments
27. Recent tobacco or nicotine product cessation within < 3 months prior to informed consent
28. Known allergies to the device components or contrast media
29. Limited life expectancy due to terminal disease
30. Currently participating in another clinical research study with an investigational drug or medical device
31. A positive COVID-19 test prior to the study procedure
32. Any condition that, in the investigator's opinion, may preclude completion of follow-up assessments throughDay 360 (e.g., a medical condition that may increase the risk associated with study participation or may interfere with interpretation of study results, inability to adhere to the visit schedule, or poor compliance with treatment regimen)

Study Procedure (Figure 1):

Participants meeting all eligibility criteria and in whom the side-to-side anastomosis duodeno-ileostomy procedure using the MagDI System (Figure 2) is attempted are considered enrolled.

The procedure (Figure 3) is an inpatient procedure, taking place in an operating room under general anaesthesia. Each Magnet is introduced orally either by swallowing [first (1 of 2) distal Magnet only] or with a flexible endoscopic catheter. Final positioning of the Magnets at the intended anastomosis site is completed using the magnetic Laparoscopic Positioning Device and optional non-magnetic laparoscopy devices, to manipulate the Magnet to the intended positions and lift the distal Magnet in the ileum to meet with the proximal Magnet in the duodenum. A non-magnetic needle with pre-attached suture and a retrieval device are included if retrieval or conversion to an enterotomy is required to position a Magnet. During the procedure, participants will be continuously monitored as per ICHT standard of care.

Upon completing the study procedure, any adverse effects, device deficiencies, protocol deviations and changes in concomitant medications will be recorded.

The study participants are expected to be discharged from the hospital within one day following the procedure, unless a longer stay is medically required as determined by the investigator. The successful placement of the MagDI System will be confirmed radiologically (abdominal x-ray) and fluoroscopically (barium or gastrographin) on Day 1. Prior to discharge, participants will meet with a dietician/nutritionist to go over the post-procedure diet.

The participants are followed up at 14, 30, 60, 90, 180, 270 and 360 days.

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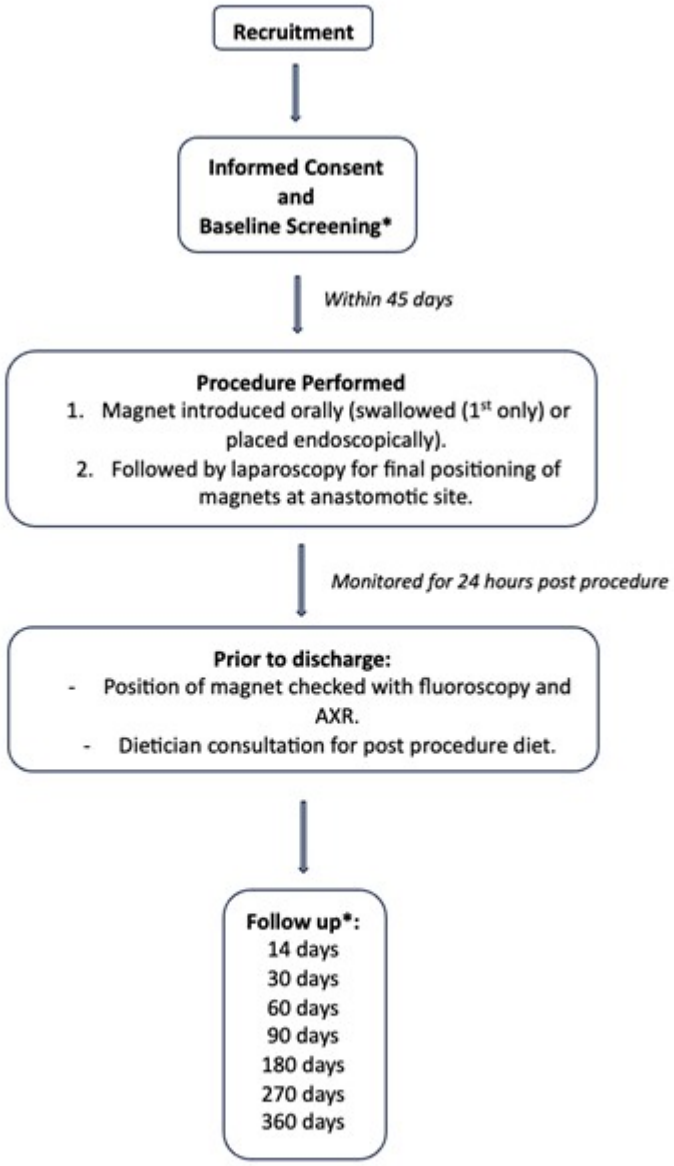

***Please see schedule of study assessments table**

Figure 1: Study Procedure Flow Chart

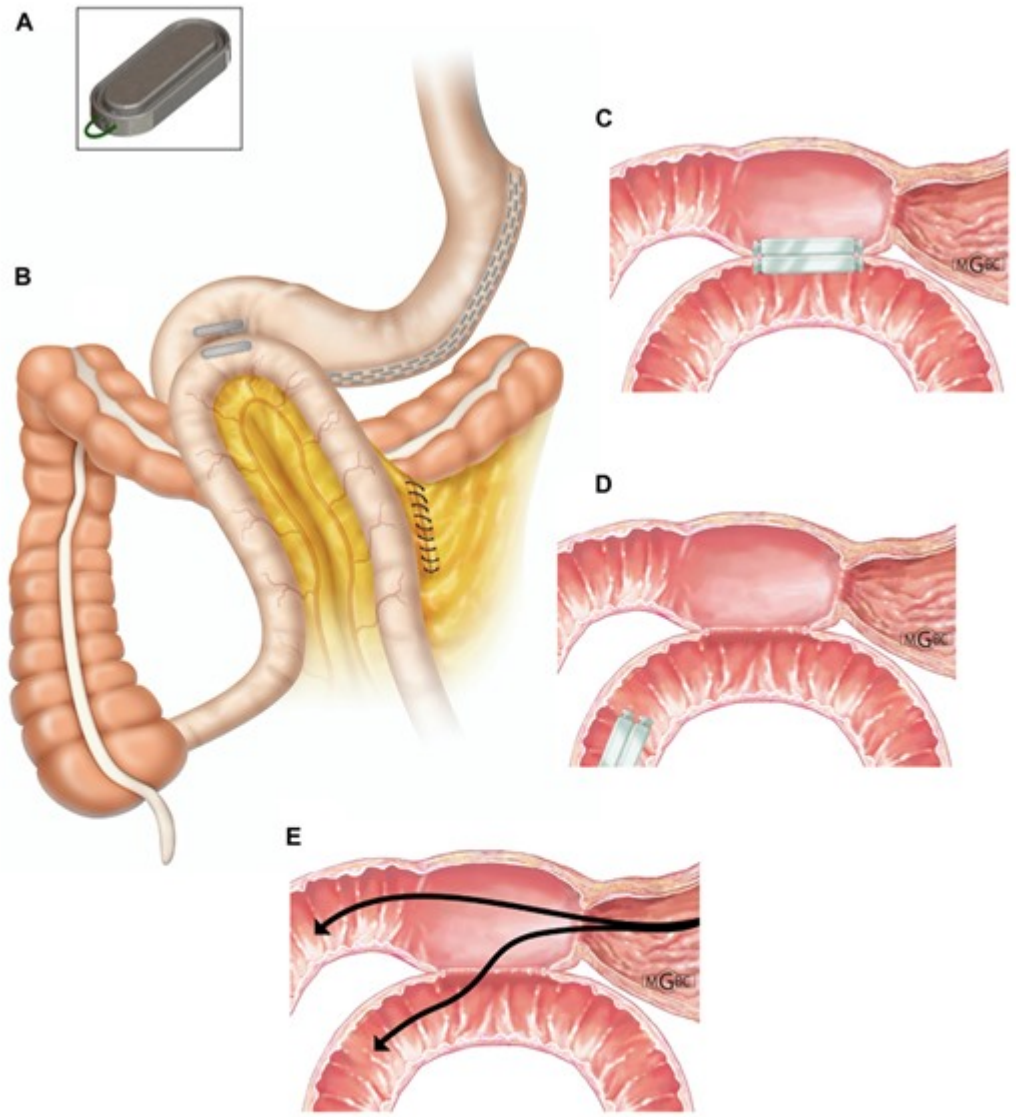

Figure 2: (A) Magnet Anastomosis System (MS) magnet. (B) After the distal MS magnet is endoscopically positioned at the ligament of Treitz and directed through the jejunal lumen to the ileum by laparoscopic magnetic positioning device assistance, it is raised anterior and latero-lateral to the post pyloric duodenum to align with the endoscopically delivered proximal MS magnet and the mesenteric space is closed. (C) In approximately 2–4 weeks, magnets fuse and the patent duodeno-ileal (DI) anastomosis is formed. (D) The magnet pair detaches and is expelled naturally. (E) Alimentary contents pass through the DI bipartition into duodenum and ileum.

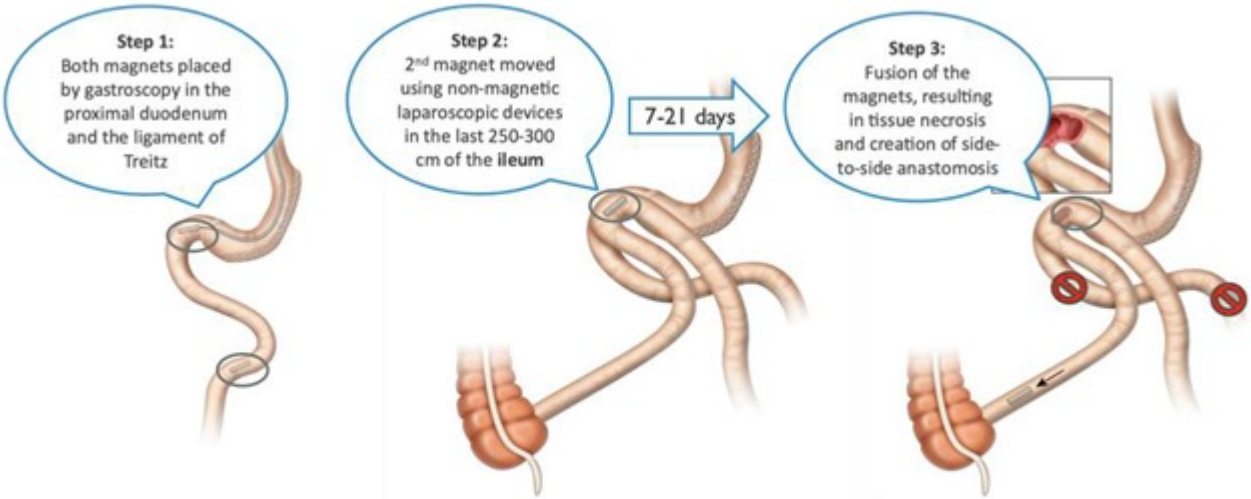

Figure 3: Procedure: Side-to-Side Compression Anastomosis using the GT Metabolic Solution Magnet System to Achieve Duodeno-Ileostomy Diversion

Assessment	Informed Consent & Screening (Baseline)	Procedure (≤ 45d post-ICF)	Day 1 (Discharge) (+ 4d)	Day 14 (Post-Op) (± 3d)	Day 30 (± 5d)	Day 60 (± 14d)	Day 90 (± 14d)	Day 180 (± 14d)	Day 270 (± 14d)	Day 360 (± 14d)
Informed Consent	X									
Inclusion / Exclusion Screening	X									
Demographics and Medical History	X									
Physical Exam Including Vital signs	X	X	X	X	X	X	X	X	X	X
Weight, Waist Circumference	X	X			X	X	X	X	X	X
Concomitant Medications	X	X	X	X	X	X	X	X	X	X
Psychologic Evaluation	X									
Surgical Clearance	X									
Clinical Laboratory Tests										
Pregnancy Test	X	X								
Blood Tests[1]	X				X	X	X	X	X	X
Imaging / Diagnostics										
Abdominal X-Ray (devices position/passing)		X[2]	X	X[3]	X[3]	X[3]	X[3]			
Fluoroscopy UGI (barium or gastrographin)			X		X		X			
Gastroduodenoscopy / Endoscopy		X			X[4]	X[4]	X[4]	X		X
Questionnaires (Surveys)										
Satisfaction Questionnaire								X		X
Other										
Device Deficiency Reporting		X	X	X	X	X				
Protocol Deviation Reporting	X	X	X	X	X	X	X	X	X	X
Adverse Event Reporting	X	X	X	X	X	X	X	X	X	X

Abbreviations: CBC, complete blood count; d, day; HbA1c, hemoglobin A1c; IC, Informed Consent; T2DM, type 2 diabetes mellitus.

[1] Blood tests include: CBC, coagulation profile *at screening*, lipid panel, liver profile, chemistry profile, thyroid function tests, HbA1c, vitamins and minerals (see **Section 6.2**.)

[2] Only for participants who opt to swallow distal Magnet (first, 1 of 2 Magnets).

[3] If Magnet passed previously, no further abdominal x-rays. Perform minimally at Day 30, and/or per PI discretion (as UVs if needed), until Magnet is no longer detectable.

[4] Gastroduodenoscopy/endoscopy performed the earlier of visit Day 30, 60, 90 following either Magnet anastomosis release (a) confirmed on x-ray, or (b) expelled/collected.

Table 2: Schedule of Study Assessments

Statistical Methods:

The primary feasibility / performance endpoint and all secondary efficacy endpoints (safety evaluation and initial efficacy) identified here are considered a priori analyses because they have been pre-specified prior to initiating enrolment. All other, analyses, if any, designed subsequent to locking the database, will be considered post-hoc analyses, and their results will be considered exploratory. Any post hoc analyses will be clearly identified in the clinical study report.

Since this is a study to evaluate feasibility/performance, safety and initial efficacy, and the sample size is small, no formal null hypotheses will be tested, and no imputation methods will be employed to replace missing or spurious data.

Numeric descriptive statistics will include the n (observed data), mean, standard deviation (SD) or standard error of the mean (SEM), median, minimum value, and maximum value. Categorical summaries will show the number and percent of subjects in the levels associated with the variable summarised and the analysis performed. Percentages will be calculated based on the total number of subjects with non-missing data for the summary. For all statistical tests a p-value of less than or equal to 0.05 on a two-sided test, or less than or equal to 0.025 on a one-sided test will be considered statistically significant, unless otherwise stated. All confidence intervals will be 95% confidence intervals unless otherwise stated.

The following is a description of the study sampling plan and a summary of the intended analyses.

Primary Endpoint:

The primary endpoint is the feasibility / performance of side-to-side anastomosis at Day 90 for duodeno-ileal diversion and will be considered feasible if it results in successful:

- Placement of the Magnets (≥90% alignment of magnets); and
- Passage of Magnets without surgical re-intervention; and
- Creation of a patent anastomosis confirmed radiologically.

The primary endpoint will be met if the feasibility/performance is confirmed in ≥80% of enrolled and treated participants. This can be summarized by the primary efficacy hypothesis:

$$H_0: p \leq 80\% \text{ vs } H_1: p > 80\%$$

Where p is the procedure success rate and 80% is the performance goal using a one-sided 0.05 test. The hypothesis test provides a statistical evaluation as previous clinical investigations using the Magnet System included the 80% cut-off for observed success as a study success criterion. Evidence from the previous device version aligns with observed results as would be needed to demonstrate success based on statistical significance for the primary study hypothesis.

Secondary Endpoints:

- Safety Measures

The incidence of AEs and SAEs from informed consent to Day 360 will be tabulated and summarized, including by severity and relationship to study device and procedure, in this clinical investigation of up to 50 participants.

The summarised incidence rates will be compared to benchmarks reported in the literature for bariatric procedures using suture and staples.

- Efficacy Endpoints

The efficacy variables will be summarised descriptively. Improvements in co-morbidities will be determined through a non-statistical analysis of vital signs and laboratory outcomes at each specified timepoint along with a systematic review of medical history and concomitant medication changes occurring throughout the study.

- Sample size

The sample size of up to 50 is chosen without a formal sample size calculation of response rate given the single-arm design and feasibility study purpose.

Ethical Considerations:

The research team are familiar with the ethical approval process. Most recently, Mr Ahmed was awarded NIHR funding for the EMBIO trial, which compared weight loss in patients undergoing left gastric artery embolisation (intervention) versus a placebo procedure.

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We are currently in the process of submitting our REC application to the HRA for approval.

Dissemination of Findings:

Findings from this study will be presented nationally and internationally and in high impact publications and NIHR reports. We will engage with the public through our PPI group, charities and social media.

6. Intellectual Property and Commercialisation

GT Metabolic Solutions, Inc. is the owner and manufacturer of The MAGNET System, all components, application procedures, and related intellectual property. The company is intending to commercialise the product.

Administrative contact details

Administrative contact name	
Administrative contact job title	
Administrative contact telephone number	
Administrative contact email address	

Research and Development office contact details

R&D contact name	
R&D contact job title	
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Uploads

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